

Statement of Exceptional Work — R. Baye

Summa of the Gap: The Complete Architecture of the Vault

889 pages | August 2026 | Spectral geometry, complexity theory, consciousness modeling, cybernetics, applied physics

What I Built

Summa of the Gap is a unified interdisciplinary treatise organized around a single principle: the gap does not close. It spans six volumes, structured as a formal research program with every claim explicitly labeled by epistemic status — [P]roven, [E]mpirical, [C]onjectured, or [I]nterpretation.

This is not a survey. It is original architecture.

Core Contributions

1. Conditional spectral gap closures (Connes program)

Derived conditions under which the spectral gap in Dirac-type operators vanishes. The gap closes only under strict operator constraints corresponding to physical boundary conditions. Otherwise, a residual gap persists — with implications for fundamental limits on spectral resolution in quantum systems.

2. Spectral approach to P vs NP

Reformulated computational complexity in spectral terms: decision problems map onto eigenvalue spectra of self-adjoint operators on Hilbert space. The P/NP distinction becomes a spectral gap condition on the verification operator — bounded spectral radius implies collapse, otherwise separation. Connects to quantum geometric tensors and phase transitions in optimization.

3. Consciousness as incomplete self-modeling

Formal proof that, under mild constraints, a system cannot fully model its own modeling. Phenomenal states are the irreducible gap between total state space and internal representational capacity. This yields a formal bound on interpretability — a sufficiently self-referential system will always contain structure opaque to its own introspection.

4. Spectral psychiatry

Applied the same operator-theoretic tools to psychiatric nosology: mental states as continuous spectra rather than discrete DSM categories. Yields testable predictions and a framework for evaluating behavior that resists categorical classification.

5. Epistemic accounting framework

Every claim in the 889-page text is tagged [P], [E], [C], or [I]. The result is a fully auditable research program with explicit confidence levels — directly applicable to alignment, evaluation, and model behavior analysis.

Why This Matters for xAI

Area	Direct Connection
Alignment & interpretability	Formal bound on self-modeling. Epistemic accounting as eval methodology.
Architecture research	Spectral operator approach to computation, phase transitions, and scaling.
Reasoning & consciousness	Functional definition of self-modeling deficits with testable predictions.
Evaluation rigor	Taxonomy of claims that supports transparent model auditing.

About Me

American citizen, 38, based in Germany. No sponsorship required. Remote-capable on US hours. Full manuscript available.

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